Isoliner · Terrain from a topographic plan: quarries, cuts, fills, dumps

A cheat sheet for the 2.16 → 2.14 → 2.03 chain · two sheets · QGIS 3.16+ · Isoliner plugin · no GRASS or SAGA

The problem

In a delivered topographic plan, areal quarries, cuts, fills, terricones and dumps **carry no contours inside**. The standard requires it, and only the outer outline is agreed with the contours by node points. Terrain built from such a plan has a gap exactly where the working is. There is nothing to fill it with except the crests and toes, which the same deliverable describes as a matter of course.

The industry requirements state plainly that terrain created automatically has to be corrected manually, agreeing it with the heights of retaining walls, slopes and fills. The chain below replaces that prescribed correction with a rule.

What is needed from the deliverable

- **Contours** with an elevation in a field or in Z. The ordinary plan layer.
- **Crests and toes** of the areal forms. In a deliverable they carry codes, for example a reinforced crest of a slope, a fill or a cut, code 62350400. No classifier holds a code for a toe, it goes as the lower contour or the line of the base.
- Spot heights, thalwegs and the water edge, if present, go into 2.03 as usual. The slope lines carry no elevations, and that is normal, step 1 gives them.

The chain

Step 1. Elevations - 2.16 Profiling of slopes. Input: the crest and toe lines plus the contour layer, adjoining tolerance 0.5 m. Output: the same lines as LineStringZ with a profile, and a separate layer of mute lines with the reason.

A contour **does not run along the crest**, contours break off at slopes. But the standard requires bringing them up to the object line with node points. Every such point carries the elevation of its contour, and the profile of the line is recovered from them.

The log prints the median and the minimum of support points per line. One point means a constant elevation along the whole line, that is, no profile came out.

Step 2. Forms - 2.14 Crests and toes into work. Input: the lines with elevations from step 1. **Leave the DEM empty**. Without it the pairs are assembled from the elevations, and the toe is the nearest line lying below the crest. Keep the path limit close to the width of the face. Output: the **Top** and **Bottom** layers sharing the **link** field, plus the unpaired with a reason.

The kind field is read tolerantly. The brow and toe values work, as do the crest codes of the classifiers, names containing the words for a crest or a toe. How every value was read is printed to the log as a table, worth checking on the very first layer.

Step 3. Terrain - 2.03 Topo2Raster. The usual inputs plus **Top of forms** and **Bottom of forms** from step 2, link field. Set the cell size explicitly.

Step 4. Check. Overlay contours from 1.04 over the result on the original ones. Outside the form they must coincide, and inside they must appear where there were none. The residual is measured by 2.09.

Why the order is what it is

The descent in 2.14 answers a single question, which way is down. That used to require a DEM, and the scenario went in a circle. To build terrain from forms you need pairs, and for pairs by descent you need terrain that does not exist yet. When the lines carry elevations, the answer is already in the data. Hence the order. Elevations first, then pairs, then terrain.

How it is computed

- **A side of a form is a set.** Any number of lines and points with one link value goes into the top or the bottom. This covers all the cases. A slope is a crest and a toe, a pit is a closed crest and a point on the floor, a ring dump is two closed lines, a ditch is two crests and a floor line.
- **No correspondence of points is sought.** For a cell the distances to both sides are taken, the weight is their ratio and the elevation is linear in the weight. An overlap is impossible by construction. The body goes in as hard nodes, the border as a barrier, and the relaxation scheme of 2.03 does not change.
- **There is no significance threshold.** Two sides give a surface, one side gives a barrier. The standard writes “expressed at the scale of the map” instead of a number, and every client has numbers of their own: 0.5 m for walls, a metre for slopes, two on a road fill.

What to read in the log

- **2.16** - contours with an elevation, how many lines received a profile, how many stayed mute, the median and the minimum of support points.
- **2.14** - the table of how the kind field was read, the number of forms, toes and crests at them, the unpaired.
- **2.03** - per form body cells, median width in cells, the mismatch of elevations where the sides converge.

Common misfires

- **“Forms narrower than two cells”.** The cell is coarser than the face. With a 7 m width and a 30 m cell the bench does not exist in the raster at any scale. Make the cell smaller, parameters will not help here.
- **The lines stayed mute.** The contours are not brought up to the crest with node points. Raise the adjoining tolerance, but carefully, because a large one starts catching foreign contours.
- **The pair did not form.** No toes lie below the crest, or the nearest one is beyond the limit. The reason is

in the attribute of the unpaired layer. Flat lines without Z are skipped in the no-DEM mode, so put them through 2.16.

- **Values of the kind field are not recognised.** The tool lists them in the log and skips them, it will not guess. Add a field of your own with brow and toe.
- **The surface is smooth where it should be benched.** There are several benches and one link for all of them. Give every bench its own link value.